Problem Set 2

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1 Introduction

This problem set was concerned with evaluating models of entry and competition. Given data from Jia (2008), I used a variety of estimation techniques to estimate the probability of entry. First, I used a probit, then a different specification of the probit, and finally an IV model to model entry simply. Next, I used an ordered probit á la Bresnahan and Reiss (1991) to model the effect of number of firms in a market on entry. Lastly, I used the Seim (2006) method to model the effect of predicted entry of a competitor on own entry.

This report is laid out as follows. Section 2 will present the profit and entry functions, section 3 will present the simple estimation results, section 4 will present the ordered probit results, and section 5 will present the Seim method results. Finally, section 6 will conclude.

2 Profit and Entry Functions

The profit function for Walmart is as follows:

The profit function for KMart is as follows:

The entry function for Walmart is as follows:

The entry function for KMart is as follows:

3 Simple Estimation

This section will present the results of the simple estimation techniques, probit under two specifications, and an IV regression. The probit specifications will differ by inclusion of competitor as a regressor. In the IV specification, I will use the log of the distance to Bentonville as an instrument for Walmart's entry in the KMart regression, and the number of small stores in the market as an instrument for KMart's entry in the Walmart regression.

3.1 Probit (No Competitor)

First, I will present the results of the probit regression without the competitor as a regressor.

$$P(Entry_j) = \Phi(\beta_0 + \beta_1 X_j) \tag{1}$$

where X_j is a vector of characteristics including log population, log retail sales per capita, percent urban, indicators for region, log distance to Bentonville (Walmart spec. only), and number of small stores

Walmart:

	Coefficient	Std. Error
Constant	-10.7057	1.037
Log Population	1.7870	0.094
Log Reatil Sales (pc)	1.4514	0.119
% Urban	0.9521	0.197
Midwest	-0.1739	0.156
Log Distance to Bentonville	-1.1322	0.085
# Small Stores	-0.0701	0.019
South	0.6751	0.153

KMart:

	Coefficient	Std. Error
Constant	-18.8344	1.239
Log Population	1.4425	0.106
Log Retail Sales (pc)	1.4964	0.135
% Urban	1.0844	0.232
Midwest	0.4124	0.145
# Small Stores	0.0055	0.019
South	0.0546	0.142

From these results, we can see that Walmart faces less baseline entry friction than Kmart, as the constant is less negative. Population effects are also stronger for Walmart, as is being in the South. The distance to Bentonville is significantly negative, implying Walmart does not want to be too close to its headquarters, which is consistent with a belief of Walmart wanting to expand into new markets. Kmart seems to have a slight gain in the Midwest and a slight boost in entry with markets who are more urban.

3.2 Probit (Competitor)

Next, I will present the results of the probit regression with the competitor as a regressor.

Walmart:

	Coefficient	Std. Error
Constant	-12.9482	1.133
Log Population	1.9728	0.101
Log Retail Sales (pc)	1.6267	0.125
% Urban	1.1248	0.202
Midwest	-0.0554	0.157
Log Distance to Bentonville	-1.1002	0.086
# of Small Stores	-0.0677	0.020
South	0.7513	0.154
KMart	-0.6573	0.113

KMart:

	Coefficient	Std. Error
Constant	-21.5571	1.340
Log Population	1.7509	0.120
Log Retail Sales (pc)	1.7245	0.141
% Urban	1.2322	0.235
Midwest	0.5770	0.149
# Small Stores	-0.0016	0.020
South	0.3305	0.150
Walmart	-0.7102	0.111

Under this specification, the presence of a competitor has a significant negative effect on entry for both Walmart and KMart, but the effect is much stronger for KMart. This is consistent with the idea that Walmart is the dominant player in the market, and KMart is more likely to be deterred by Walmart's presence than vice versa.

3.3 IV Regression

Next, I will present the results of the IV regression. The results are as follows:

$$P(Entry_j) = \Phi(\beta_0 + \beta_1 X_j + \beta_2 Competitor_k) \tag{2}$$

where X_j is a vector of characteristics including log population, log retail sales per capita, percent urban, indicators for region, log distance to Bentonville (Walmart spec. only), and number of small stores. The competitor variable is instrumented as described above.

$$P(Entry_i) = \Phi(\beta_0 + \beta_1 X_i + \beta_2 Com\hat{p}etitor_k) \tag{3}$$

Walmart:

	Coefficient	Std. Error
Constant	-10.6671	nan
Log Population	1.7870	0.094
Log Retail Sales (pc)	1.4514	0.119
% Urban	0.9521	0.197
Midwest	-0.1739	0.156
Log Distance to Bentonville	-0.0459	nan
South	0.6751	0.153
$\hat{KMart}$	-0.6425	nan

KMart:

	Coefficient	Std. Error
Constant	-17.4435	1.260
Log Population	1.4670	0.108
Log Retail Sales (pc)	1.4752	0.137
% Urban	1.2727	0.239
Midwest	1.1634	0.173
South	0.7797	0.169
# Small Stores	0.0222	0.020
$\hat{Walmart}$	-4.5786	0.581

The IV results show that Walmart's predicted entry has a massively negative effect on KMart's entry probability, implying that a) Walmart's entry is a strong deterrent to KMart's entry, and b) the OLS results were biasing the results downwards. For Walmart, KMart's predicted entry has a negative effect, but there is likely a misspecification in instrument as standard errors are not available. To be diagnosed. However, the general magnitude is consistent with the OLS results, implying that KMart is less of a deterrent to Walmart.

4 Ordered Probit

4.1 Specification 1

In the first specification of the ordered probit, we group Walmart and Kmart together, and model the number of firms in the market as a function of the same characteristics as above. The results are as follows:

	Coefficient	Std. error
Log Population	1.9054	0.080
Log Retail Sales (pc)	1.6348	0.101
% Urban	1.2092	0.168
Midwest	0.4985	0.132
Log Distance to Bentonville	-0.3384	0.056
# Small Stores	-0.0451	0.015
South	0.8400	0.131
Effect of $0 \to 1$	17.8011	0.937
Effect of $1 \to 2$	0.8412	0.032

Here, we see that being the first entrant has a significant positive effect on entry, but the second entrant is a fraction of the effect, though still significant. This result lends itself to the idea of first mover advantages, as the first entrant is able to capture a large portion of the market. We still see a negative effect of distance to Bentonville, but the effect is smaller than in the simple probit, which is consistent with the idea that distance to Bentonville is more of a deterrent for Walmart than KMart. We also see that as there are more small stores, there is a deterrance to entry, though this effect is small.

4.2 Specification 2

Here, we group all firms together, modeling number of firms in the market as a function of the same characteristics as above. The results are as follows:

	Coefficient	Std Error
Log Population	1.7589	0.073
Log Retail Sales (pc)	1.2985	0.085
% Urban	1.1849	0.152
Midwest	0.5019	0.124
Log Distance to Bentonville	-0.2866	0.053
# Small Stores	2.7925	0.071
South	0.7867	0.122
$0 \rightarrow 1$ **Firm**	14.0616	0.801
$1 \rightarrow 2$ **Firms**	1.2129	0.047
$2 \rightarrow 3$ **Firms**	1.0088	0.042
$3 \rightarrow 4$ **Firms**	1.0971	0.041
$4 \rightarrow 5$ **Firms**	1.0198	0.045
$5 \rightarrow 6$ **Firms**	1.0504	0.049
$6 \rightarrow 7$ **Firms**	1.0265	0.053
$7 \rightarrow 8$ **Firms**	0.9840	0.064
$8 \rightarrow 9$ **Firms**	0.8639	0.080
$9 \rightarrow 10$ **Firms**	1.2793	0.068
$10 \rightarrow 11$ **Firms**	0.8734	0.119
$11 \rightarrow 12$ **Firms**	1.0653	0.109
$12 \rightarrow 13$ **Firms**	0.9233	0.121

The first mover advantage is still present, but to a lesser degree. Second mover advantage has also increased. Interestingly, after being the first mover, the number of firms in the market has a relatively stable effect on entry, implying that only first mover advantage is important, and after that, the number of firms in the market does not have a major effect on entry. Distance to Bentonvill is still negative, but less so than in the above. Small stores are now positive, which makes sense given small stores are more likely to enter into markets where other small firms are present.

5 Seim Method

In this section, I will present the results of the Seim method. The results are as follows:

Variable	Walmart Coefficient	KMart Coefficient
Log Population	6.68	1.25
Log Retail Sales (pc)	0.11	-1.03
Percent Urban	8.95	3.10
Midwest	-1.63	0.10
Log Distance to Bentonville	-3.28	0.17
Number of Small Stores	0.06	0.10
South	-1.55	-0.82
Predicted KMart Entry	1.15	–
Preidcted Walmart Entry	–	-14.38

6 Conclusion

In conclusion, I have presented some data facts, estimation results, and a regime comparison for the problem set. I will have to go back and check my code for the regime comparison, as the results are not what I expected. However, I still feel that this exercise was very useful in a few dimensions. Firstly, it was a good refresher on the different estiamtion techniques, and the biases that can result from misspecified methods of computation. Secondly, it was good for thinking of pricing games and how to derive the FOC's for the different regimes. Finally, the data exploration exercise was good as I am trying to learn Python and its plotting libraries.

A Code

A.1 Main Code

```
import numpy as np
import scipy as sp
import statsmodels.api as sm
import pandas as pd
from prettytable import PrettyTable
from statsmodels.miscmodels.ordinal_model import OrderedModel


"""
Problem␣Set␣2:

Author:␣Tate␣Mason
Due:␣04-23-2026
Class:␣IO␣2



␣␣*␣Helper␣files:
␣␣␣␣-␣PS2c.py:␣Contains␣the␣implementation␣of␣the␣Seim␣approach␣for␣estimating␣entry␣
    decisions␣in␣a␣duopoly␣setting.

␣␣*␣Libraries␣used:
␣␣␣␣-␣numpy
␣␣␣␣-␣scipy
␣␣␣␣-␣statsmodels
␣␣␣␣-␣pandas
␣␣␣␣-␣prettytable
"""

# Read Data

xmat_cols = [
    'county_id', 'log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
        log_dist_benton', 'south',
    'kmart', 'walmart', 'n_small_stores', 'dw_kmart_out', 'dw_walmart_out', 'col13', '
        col14', 'col15'
]

XMat = pd.read_csv('../Data/XMat.out', sep=r'\s+', header=None, names=xmat_cols)
print(XMat.head())

# Helper function to save LaTeX tables
import os
os.makedirs('../Tables', exist_ok=True)
def save_tex_table(latex_str, filename):
    with open(f'../Tables/{filename}.tex', 'w') as f:
```

```
        f.write(latex_str)

#=== Problem 2a ===#

"""
Here␣we␣will␣use␣a␣simple␣probit␣under␣different␣specifications␣to␣estimate␣the␣entry␣
    decision␣of␣a␣firm

-␣Data:
␣␣␣␣-␣XMat.out:␣County␣level␣data␣on␣various␣store␣level␣controls.
"""

from PS2a import *

#=== Part A ===#

print("===␣Part␣A␣===")

lhs_wm = XMat['walmart']
rhs_wm = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    log_dist_benton', 'n_small_stores', 'south']]

wm_probit = sm.Probit(lhs_wm, sm.add_constant(rhs_wm)).fit()
save_tex_table(wm_probit.summary().as_latex(), 'wm_probit_spec1')

lhs_km = XMat['kmart']
rhs_km = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'south']]

km_probit = sm.Probit(lhs_km, sm.add_constant(rhs_km)).fit()
save_tex_table(km_probit.summary().as_latex(), 'km_probit_spec1')

#=== Part B ===#

print("===␣Part␣B␣===")

lhs_wm_comp = XMat['walmart']
rhs_wm_comp = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    log_dist_benton', 'n_small_stores', 'south', 'kmart']]

wm_comp_probit = sm.Probit(lhs_wm_comp, sm.add_constant(rhs_wm_comp)).fit()
save_tex_table(wm_comp_probit.summary().as_latex(), 'wm_probit_spec2')

lhs_km_comp = XMat['kmart']
rhs_km_comp = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'south', 'walmart']]

km_comp_probit = sm.Probit(lhs_km_comp, sm.add_constant(rhs_km_comp)).fit()
```

```
save_tex_table(km_comp_probit.summary().as_latex(), 'km_probit_spec2')

# The biggest weakness of using the number of competitor stores as a regressor is the
    almost assured endogeneity. The presence of a competitor store is likely correlated
     with unobserved factors that also affect the entry decision, such as local market
    conditions, consumer preferences, or the strategic behavior of firms.
# For example, if a county has favorable market conditions for retail entry (e.g.,
    high population density, strong consumer demand), both Walmart and Kmart may be
    more likely to enter, leading to a positive correlation between the presence of one
     firm and the other. This endogeneity can bias the estimated coefficients and lead
    to incorrect inferences about the competitive effects on entry decisions.

#=== Part C ===#
print("=== Part C ===")

lhs_km_iv = XMat['kmart']
rhs_km_iv = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'south', 'walmart']]
instrument_km_iv = 'log_dist_benton'

# First stage regression
first_stage_km_iv = sm.OLS(XMat['walmart'], sm.add_constant(XMat[instrument_km_iv])).
    fit()
XMat['predicted_walmart'] = first_stage_km_iv.predict(sm.add_constant(XMat[
    instrument_km_iv]))

# Second stage regression

rhs_km_iv2 = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', 'south',
    'n_small_stores', 'predicted_walmart']]
model_km_iv = sm.Probit(lhs_km_iv, sm.add_constant(rhs_km_iv2))
results_km_iv = model_km_iv.fit()
print(results_km_iv.summary())
save_tex_table(results_km_iv.summary().as_latex(), 'km_iv_probit')

lhs_wm_iv = XMat['walmart']
rhs_wm_iv = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'log_dist_benton', 'south', 'kmart']]
instrument_wm_iv = 'n_small_stores'

# First stage regression
first_stage_wm_iv = sm.OLS(XMat['kmart'], sm.add_constant(XMat[instrument_wm_iv])).fit
    ()
XMat['predicted_kmart'] = first_stage_wm_iv.predict(sm.add_constant(XMat[
    instrument_wm_iv]))

# Second stage regression
```

```python
rhs_wm_iv2 = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'log_dist_benton', 'south', 'predicted_kmart']]
model_wm_iv = sm.Probit(lhs_wm_iv, sm.add_constant(rhs_wm_iv2))
results_wm_iv = model_wm_iv.fit()
print(results_wm_iv.summary())
save_tex_table(results_wm_iv.summary().as_latex(), 'wm_iv_probit')

# Kmart IV results suggest that coeff on pred. WM is significant and negative. This
    implies that as distance from benton co. increases, the probability of Walmart
    entry increases, which in turn reduces the probability of Kmart entry.
# This makes sense, as WM would be more likely to enter as they stray from their home
    county. The exclusion restriction seems plausible, as distance from Benton Co.
    should not directly affect Kmart entry decisions, except through its effect on
    Walmart's entry.
# Relevance condition is satisfied as distance from Benton Co. is likely correlated
    with Walmart's entry decision, given that Benton Co. is Walmart's home county and a
     central location for their operations.
#
# WM IV coef on pred. KM is significant and negative, suggesting that as the number of
     small stores increases, the probability of Kmart entry increases, which in turn
    reduces the probability of Walmart entry. This also makes sense, as a higher number
     of small stores may indicate a more competitive retail environment, making it less
     attractive for Walmart to enter.
# The exclusion restriction seems implausible, as the number of small stores could
    directly affect Walmart's entry decision, independent of Kmart's entry. For example
    , a high number of small stores could indicate a saturated market, which would
    deter Walmart from entering regardless of Kmart's presence.
# The relevance condition is satisfied as the number of small stores is likely
    correlated with Kmart's entry decision, as Kmart may be more likely to enter
    markets with fewer small stores where they can capture more market share.

#=== Bresnahan-Reiss Approach ===#

# Ordered Probit

# Spec 1: LHS = number of large players, RHS = market controls

controls = ['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', 'log_dist_benton
    ', 'n_small_stores', 'south']

XMat['n_large'] = XMat['kmart'].astype(int) + XMat['walmart'].astype(int)

res_spec1 = OrderedModel(
    XMat['n_large'], XMat[controls], distr='probit'
).fit(method='BFGS')

print(res_spec1.summary())
save_tex_table(res_spec1.summary().as_latex(), 'ordered_probit_spec1')
```

```python
# Spec 2: LHS = number of players (large + small), RHS = market controls

XMat['n_total'] = XMat['n_large'] + XMat['n_small_stores']

res_spec2 = OrderedModel(
    XMat['n_total'], XMat[controls], distr='probit'
).fit(method='BFGS')

print(res_spec2.summary())
save_tex_table(res_spec2.summary().as_latex(), 'ordered_probit_spec2')

#=== Seim Approach ===#

from PS2c import *


XW = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', 'log_dist_benton'
    , 'n_small_stores', 'south']].values
XK = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', 'n_small_stores',
     'log_dist_benton', 'south']].values

yW = XMat['walmart'].values
yK = XMat['kmart'].values

K = XW.shape[1]

theta0 = np.zeros(2*K + 2)

res_seim = minimize(log_lik, theta0, args=(XW, XK, yW, yK), method='BFGS')

theta_hat = res_seim.x
betaW_hat = theta_hat[:K]
deltaW_hat = theta_hat[K]
betaK_hat = theta_hat[K+1:2*K+1]
deltaK_hat = theta_hat[-1]

res_table = PrettyTable()
res_table.field_names = ["Parameter", "Estimate"]
for i in range(K):
    res_table.add_row([f"betaW_{i}", betaW_hat[i]])
res_table.add_row(["deltaW", deltaW_hat])
for i in range(K):
    res_table.add_row([f"betaK_{i}", betaK_hat[i]])
res_table.add_row(["deltaK", deltaK_hat])
print(res_table)
param_names = [f'betaW_{i}' for i in range(K)] + ['deltaW'] + \
              [f'betaK_{i}' for i in range(K)] + ['deltaK']
```

```
seim_df = pd.DataFrame({
    'Parameter': param_names,
    'Estimate': theta_hat
})

save_tex_table(
    seim_df.to_latex(index=False, float_format='%.4f', caption='Seim␣NFP␣Estimates',
        label='tab:seim'),
    'seim_results'
)
```

A.2 Seim Code

```
import numpy as np
import scipy as sp
from scipy.special import expit
from scipy.optimize import minimize

def solve_eq(xW, xK, theta, eps=1e-6, max_iter=1000):
    betaW = theta[:xW.shape[0]]
    deltaW = theta[xW.shape[0]]

    betaK = theta[xW.shape[0]+1 : xW.shape[0] + 1 + xK.shape[0]]
    deltaK = theta[-1]

    PW, PK = 0.5, 0.5

    for _ in range(max_iter):
        PW_new = expit(xW@betaW + deltaW*PK)
        PK_new = expit(xK@betaK + deltaK*PW)

        if max(abs(np.log(PW_new / PW)), abs(np.log(PK_new / PK))) <= eps:
            return PW_new, PK_new
        PW, PK = PW_new, PK_new

    return PW, PK

def log_lik(theta, XW, XK, yW, yK):
    M = XW.shape[0]
    ll = 0.0

    for m in range(M):
        PW, PK = solve_eq(XW[m], XK[m], theta)

        PW = np.clip(PW, 1e-10, 1 - 1e-10) # to avoid log(0)
        PK = np.clip(PK, 1e-10, 1 - 1e-10) # to avoid log(0)

        ll += yW[m]*np.log(PW) + (1 - yW[m])*np.log(1 - PW)
```

```
        ll += yK[m]*np.log(PK) + (1 - yK[m])*np.log(1 - PK)
    return -ll
```